

Badre Mhiouah

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Quantitative Finance & Data Science

SUMMARY

Quantitative developer with 1.5 years of experience in derivatives trading—including XVA and gap-risk analytics—and AI-driven risk solutions at a bulge-bracket bank. Passionate about multi-asset and structured-product quant research, I am eager to contribute to AXA Investment Managers' evolving quantitative teams, leveraging my background in statistical modelling, machine learning, and production-grade Python development. I am aware of the current strategic combination between AXA IM and BNPParibas Asset Management and look forward to supporting the integrated quant capabilities.

EXPERIENCE

Barclays : Trader intern

Aug.2023 – Jan. 2024

Cross-asset derivatives trading desk

Paris & London

- Developed a **trading strategy aimed at hedging own credit risk**, resulting in a **7% boost in profitability** and a **5% decrease in P&L volatility**. (Bloomberg, Python, Machine learning)
- Improved **risk management instruments** to pinpoint counterparties with significant impacts on each XVA by country, currency and greeks. (Excel, VBA, API)
- Modeled and calculated the effects of **gap risk** on both credit and own credit risks for the portfolio of the bank.
- Developed **reporting tools** to regularly analyze gap risk, facilitating the identification of its key drivers.
- Collected and cleaned **interest rate data** for major currencies, automated data processing, and developed optimal **neural network models for yield curve prediction** with backtesting and hyperparameter tuning.

Societe Generale Corporate and Investment Banking : Data Scientist

Jun.2025 – Jul.2026

General Inspection

Hong Kong

- Partnered with **non-technical stakeholders** to identify **time-consuming, repetitive audit activities** and assess opportunities for automation, translating business needs into **data and AI use cases**.
- Designed and prototyped **LLM-based solutions** using **prompt engineering and LLM APIs** to automate and accelerate audit workflows, including the enrichment, structuring, and analysis of **operational loss data**.
- Developed a **Power BI platform** combining automated insights and **multi-level risk reporting**, enabling auditors to reduce manual analysis and gain faster access to **actionable risk information**.
- Analyzed **large-scale trading datasets** from Delta One activities to identify **anomalous patterns, risk concentrations, and control weaknesses** across equity and index derivative products, supporting auditors in risk-focused investigations.
- Performed **forensic analyses of trading and market data** to identify potential indicators of **market manipulation**.
- Delivered **Python and Power BI training** to auditors, enabling **self-service analytics** and facilitating the adoption of data-driven approaches within the IGAD teams.

PERSONAL PROJECTS (AVAILABLE ON MY GITHUB)

Financial News Sentiment & Market Analysis Pipeline

Aug. 2026 – Present

Python, Hugging Face, FinBERT, FastAPI, PostgreSQL, Docker

- **Goal:** Building an end-to-end AI pipeline to extract sentiment from financial news, transform it into quantitative signals, and evaluate its relationship with subsequent **stock market movements**.
- **AI & Data Pipeline:** Collecting financial news and historical market data, processing and storing them in a database, and using **FinBERT** for domain-specific sentiment analysis and continuous sentiment scoring.
- **Quantitative Analysis:** Aggregating sentiment into trading signals and implementing simple **trading strategies and backtests** to evaluate predictive power, returns, and risk against baseline approaches.
- **AI Engineering:** Developing a modular Python application and progressively exposing its inference and analysis capabilities through **FastAPI**, with **PostgreSQL** and **Docker** planned for scalable data management and reproducible deployment.

Jane Street Puzzle – August 2026

Aug. 2026

Mathematics, Python, LLMs, Cursor

- **AI-Assisted Reasoning:** Using multiple frontier LLMs and **Cursor** to explore mathematical approaches, generate code and challenge proposed solutions.
- **Computational Validation:** Developing Python simulations and visualizations to test mathematical intuitions and validate analytical results.

EDUCATION

CentraleSupélec

Sep. 2020 – Apr. 2023

Engineering Degree, Major in Mathematical Modelling & Data Science, Minor in Financial Mathematics

Paris, France

- Relevant coursework: Mathematics, Statistics, Machine Learning, Optimization, Stochastic Models, Financial Modeling, Risk Management, Derivatives, Fixed Income, Economics, Negotiation course, Data structures, Algorithms & Complexity.

Université Côte d'Azur

Sep 2017 – Apr. 2020

Bachelor's degree in Advanced Mathematics

Nice, France

- Main courses: Topology, Complex analysis, Numerical analysis, Differential calculus, Algebra & Geometry.

EXTRACURRICULAR ACTIVITIES

Gap Year – Independent Travel (Southeast Asia)	Jan. 2024 – Jan. 2025
<i>Independent Travel</i>	<i>Southeast Asia</i>
• Undertook a year of independent travel across Southeast Asia, gaining cross-cultural experience and adaptability while navigating diverse environments and logistical challenges. • Developed strong self-management, problem-solving, and communication skills through extended travel in countries including Thailand, Vietnam, Cambodia, Indonesia. • Exposure to different cultural, economic, and social contexts strengthened intercultural awareness and resilience.	

SKILLS & INTERESTS

Agentic AI:	LLM Applications, Agentic AI, LangGraph, Tool Calling, MCP, Multi-Agent Systems, Context Engineering, LangChain, LLM APIs
RAG & Evaluation:	RAG, Embeddings, Vector Search, Hybrid Search, Reranking, LLM Evaluation, Ragas, DeepEval, Observability
AI Engineering:	Python, SQL, FastAPI, REST APIs, Docker, Git, CI/CD, AWS/GCP, Monitoring.
ML & Fine-Tuning:	PyTorch, Hugging Face, Transformers, LoRA/QLoRA, PEFT, Deep Learning, Data mining, Inference, Prediction
Data:	Pandas, NumPy, Scikit-learn, Experiment Design, Statistical Testing
Languages:	French, Arabic, English (fluent); Spanish (intermediate); Mandarin (elementary)